Introduction to Network Security and Cryptography

Aim:

To study the fundamental concepts of network security and cryptography, including types of security, OSI security architecture, the network security model, and basic cryptographic concepts.

Security:

Security is the process of protecting systems, networks, information, and resources from unauthorized access, use, disclosure, modification, disruption, or destruction.
The primary objectives of security are commonly expressed through the CIA Triad:

- ***Confidentiality*** – Preventing unauthorized disclosure of information.
- ***Integrity*** – Ensuring that information is not improperly modified or destroyed.
- ***Availability*** – Ensuring that authorized users can access information and services when required.

Types of Security –

- **Physical Security:** It refers to the protection of physical assets, facilities, hardware, and infrastructure from unauthorized physical access, damage, theft, or destruction.
  Examples: CCTV surveillance, biometric authentication, etc.

- **Communication Security:** It is the protection of information while it is being transmitted over communication systems against unauthorized interception, disclosure, modification, or disruption.
  Examples: Encryption of messages, communication protocols, digital signatures

- **Network Security:** It is the protection of computer networks and their resources from unauthorized access, misuse, attacks, disruption, or destruction.
  It uses a combination of hardware, software, protocols, and security mechanisms.
  Examples: Firewalls, network encryption, secure protocols such as HTTPS and SSH

- **Information Security:** It is the protection of information and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction.
  It aims to preserve:
  - **Confidentiality**
  - **Integrity**
  - **Availability**

  Information security covers information in digital as well as physical form.

 Signature of the Faculty.....

OSI Security Architecture

The OSI Security Architecture provides a systematic framework for describing security attacks, security services, and security mechanisms required for protecting networked systems.

It classifies security considerations into three major areas:

1. Security Attacks
2. Security Services
3. Security Mechanisms

Security Attacks

A **security attack** is any action that compromises the security of information or information systems.

Security attacks are broadly classified into:

A. Passive Attacks

A passive attack attempts to obtain information without affecting the operation of the system or modifying the data.

Examples: Release of message contents, traffic analysis

B. Active Attacks

An active attack involves modification, disruption, or fabrication of information or system operations.

Examples: Masquerade, replay, modification of messages, Denial of Service (DoS)

Security Threat

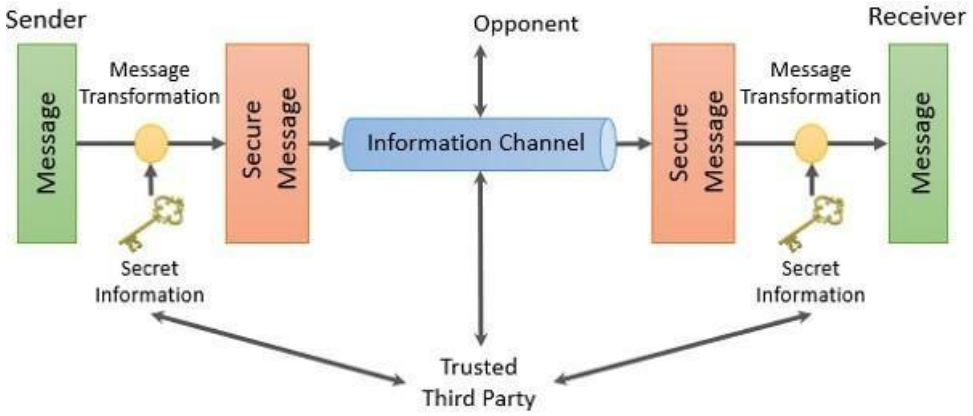
- A **security threat** is a potential circumstance, event, or action that may exploit a vulnerability and cause harm to a system or its information.
- Examples include unauthorized access, malware, eavesdropping, data modification, Denial of service, identity theft
- A threat represents **potential danger**, whereas an attack is an **actual attempt or action** against a system.

Security Mechanism

- A **security mechanism** is a process or technique designed to detect, prevent, or recover from security attacks.
- Common security mechanisms include: Encipherment, Digital Signature, Access Control, Data Integrity Mechanisms, etc.

Model for Network Security

A network security model describes how two communicating parties can securely exchange information over an insecure network in the presence of an attacker.



Network Security Model

Basic Components

1. **Sender** – The entity that generates the original message.
2. **Receiver** – The intended entity that receives the message.
3. **Message** – The information being communicated.
4. **Security Transformation** – Encryption, authentication, hashing, digital signature,
5. **Secret Information/Key** – Information used by the security transformation.
6. **Insecure Network** – The communication medium through which data travels.
7. **Attacker** – An entity attempting to intercept, fabricate, or disrupt communication.
8. **Trusted Third Party** – May be used for distributing keys or establishing trust.

Cryptography

Cryptography is the science of securing information by transforming it into a form that prevents unauthorized parties from understanding or manipulating it.
It provides security properties such as:

- Confidentiality
- Integrity
- Authentication
- Non-repudiation

Plaintext

It is the original, readable message or data that is given as input to an encryption algorithm.
Example – Hello

Ciphertext

It is the transformed, unreadable form of plaintext produced by an encryption algorithm. The ciphertext should not reveal the original message to an unauthorized party.

Example – Hello -> Khoor

Encryption

Encryption is the process of converting plaintext into ciphertext using an encryption algorithm and a key.

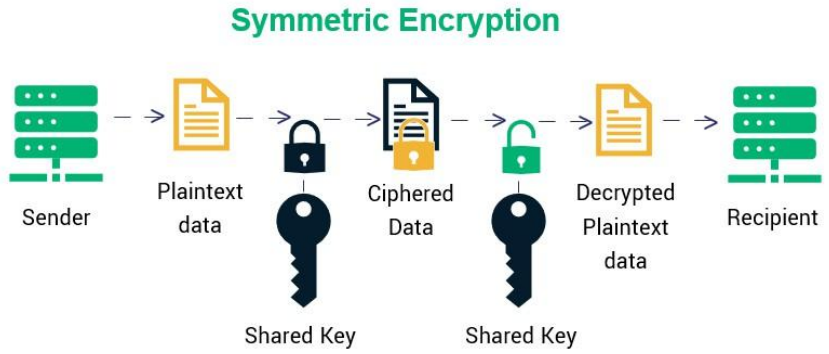
Plain text + Key = Cipher Text

Types of Encryption

Symmetric-Key Encryption

Symmetric encryption is a cryptographic technique in which the same secret key, or closely related keys, is used for both encryption and decryption.

Examples – AES, DES, 3DES



Advantages

- Fast
- Efficient for large amounts of data
- Requires relatively less computational power

Limitation

The secret key must be securely shared between the communicating parties.

Asymmetric encryption

It is also called public-key cryptography, uses a pair of mathematically related keys:

- **Public key** – Can be freely distributed.
- **Private key** – Must be kept secret.

The keys perform complementary cryptographic operations depending on the application.

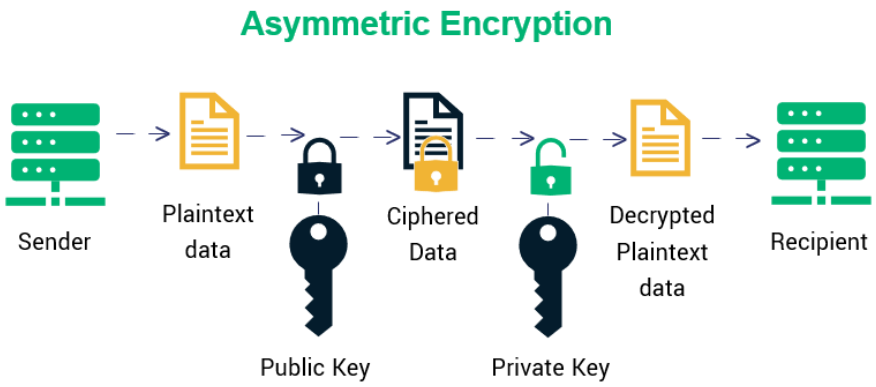
Examples – RSA, ElGamal, ECC, etc.

Advantages

- Solves the key-distribution problem associated with symmetric encryption.
- Can support digital signatures and authentication.

Limitations

- Computationally more expensive than symmetric encryption.
- Generally slower for large amounts of data.



Decryption

Decryption is the process of converting ciphertext back into the original plaintext using an appropriate decryption algorithm and key.

Cryptanalysis

Cryptanalysis is the study and practice of analyzing cryptographic systems with the objective of discovering weaknesses, recovering plaintext, determining cryptographic keys, or otherwise defeating the intended security of a cryptosystem.

Result

The fundamental concepts of network security, security attacks and mechanisms, the OSI security architecture, network security models, and cryptographic techniques including symmetric and asymmetric encryption were studied successfully.

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